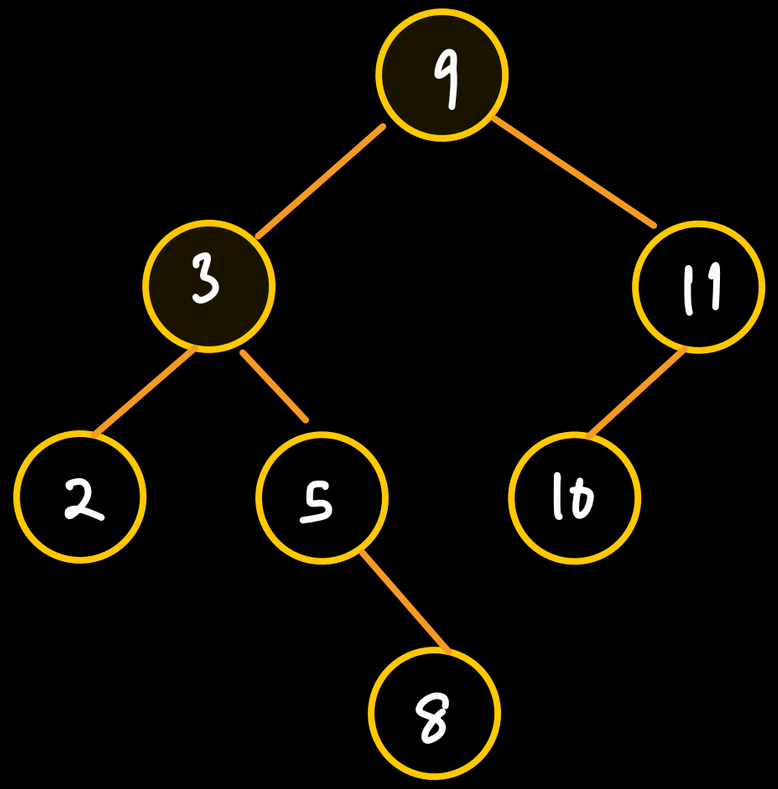
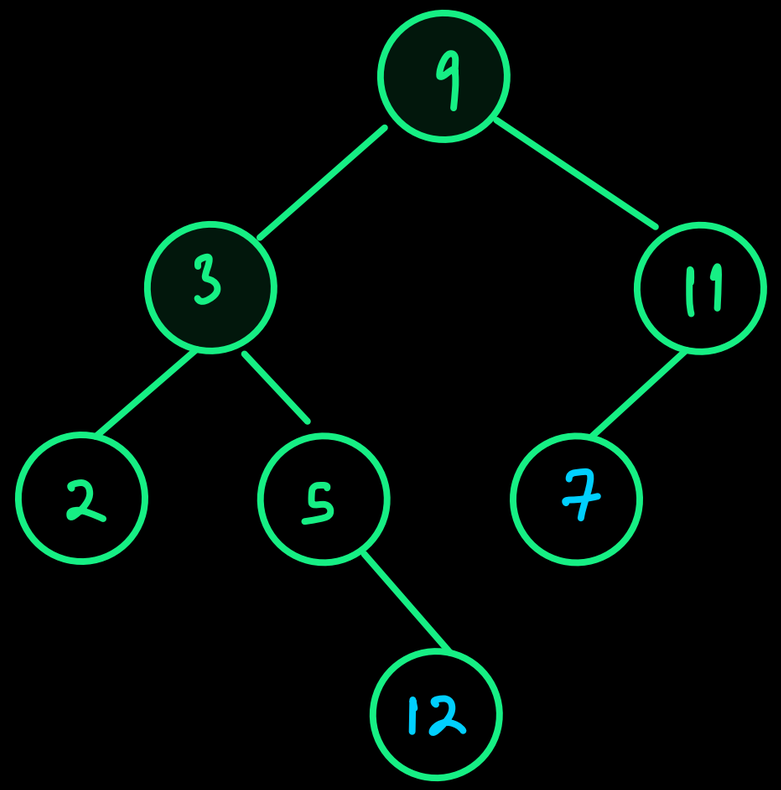
Binary
Search Trees

What is BST



Valid BST



Invalid BST

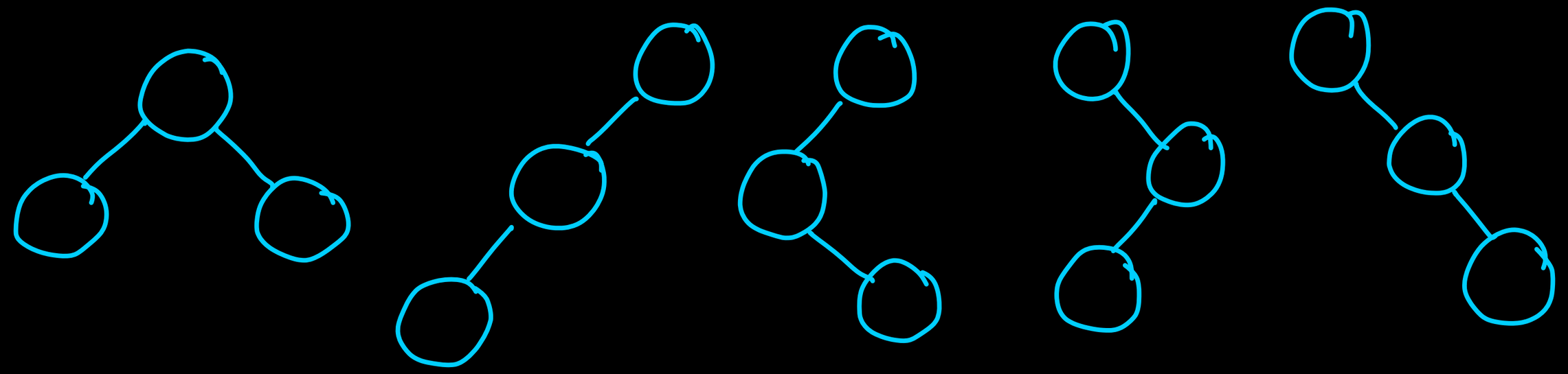
Ques: Find the Minimum & Maximum element in a Binary Search Tree

min →

```
Node temp = root
while (temp.left != null)
    temp = temp.left
return temp.val
```

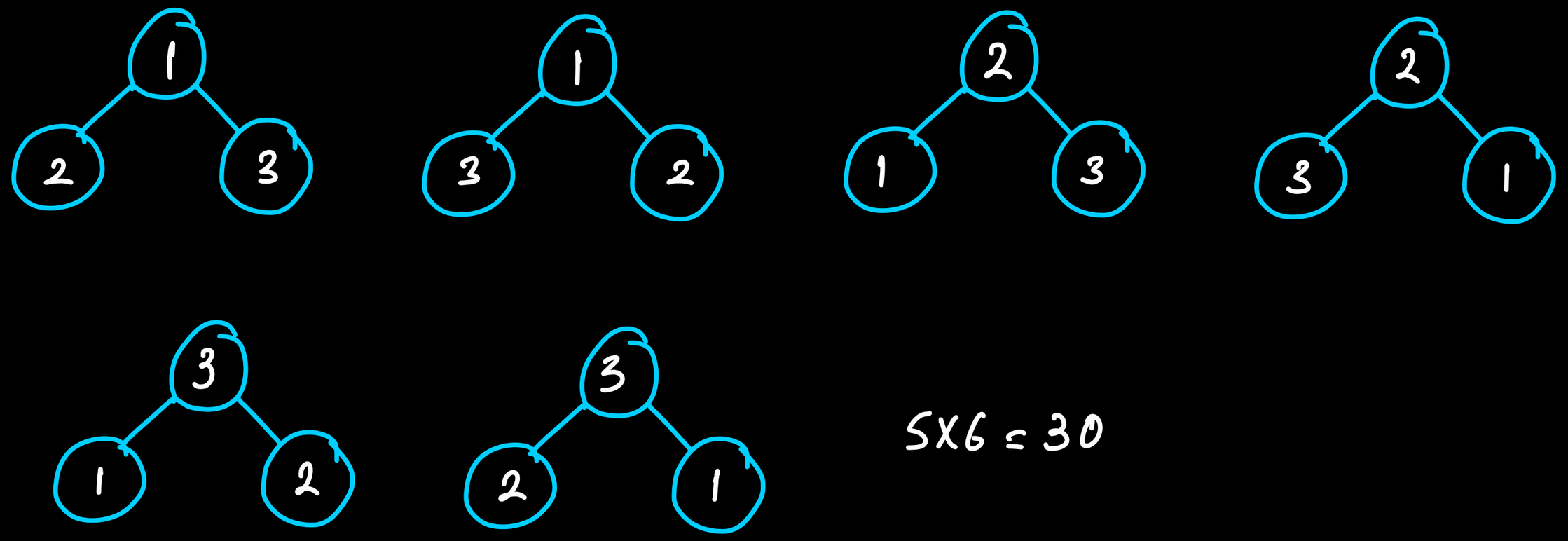
Concept Builder

- You have 3 nodes. How many unique structured Binary Trees can be formed?



Concept Builder

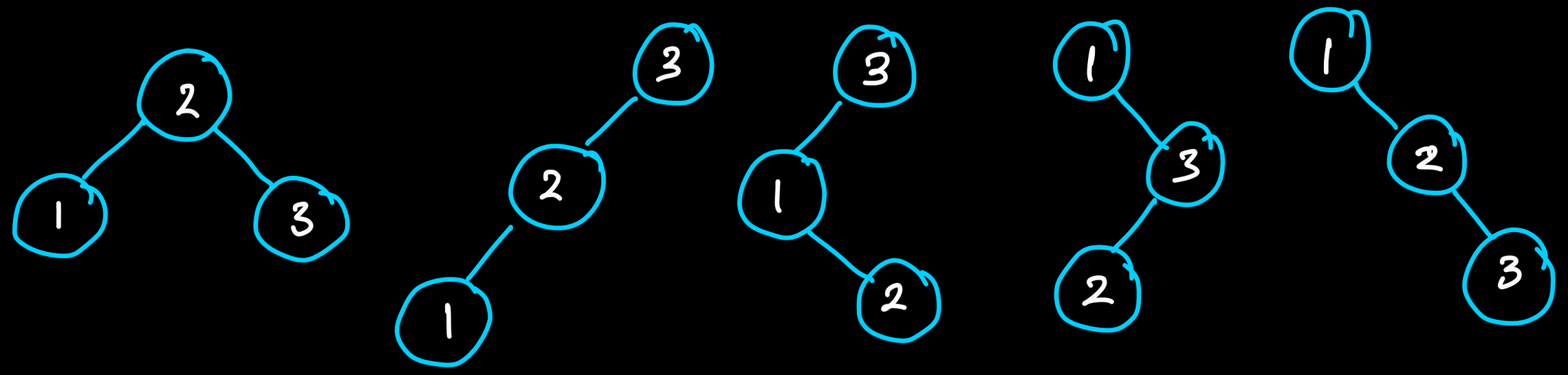
- You have 3 nodes with different values. How many unique Binary Trees can be formed? → 30



$5 \times 6 = 30$

Concept Builder

- You have 3 nodes with different values. How many unique Binary Search Trees can be formed?



No. of BST's = No. of structures

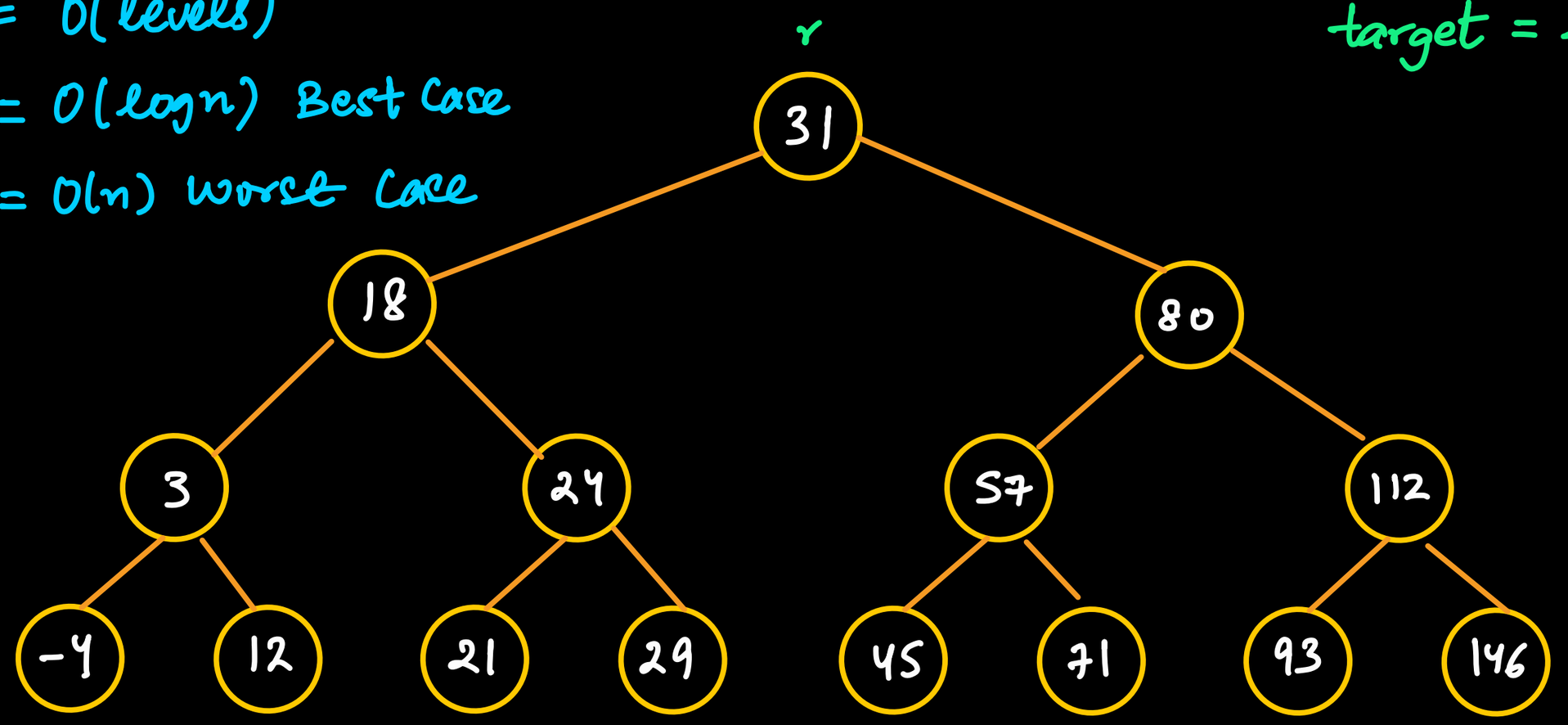
Ques: Search in a Binary Search Tree

T.C. = O(levels)

= O(logn) Best Case

= O(n) worst case

target = 21



HW: Range Sum of Binary Search Tree

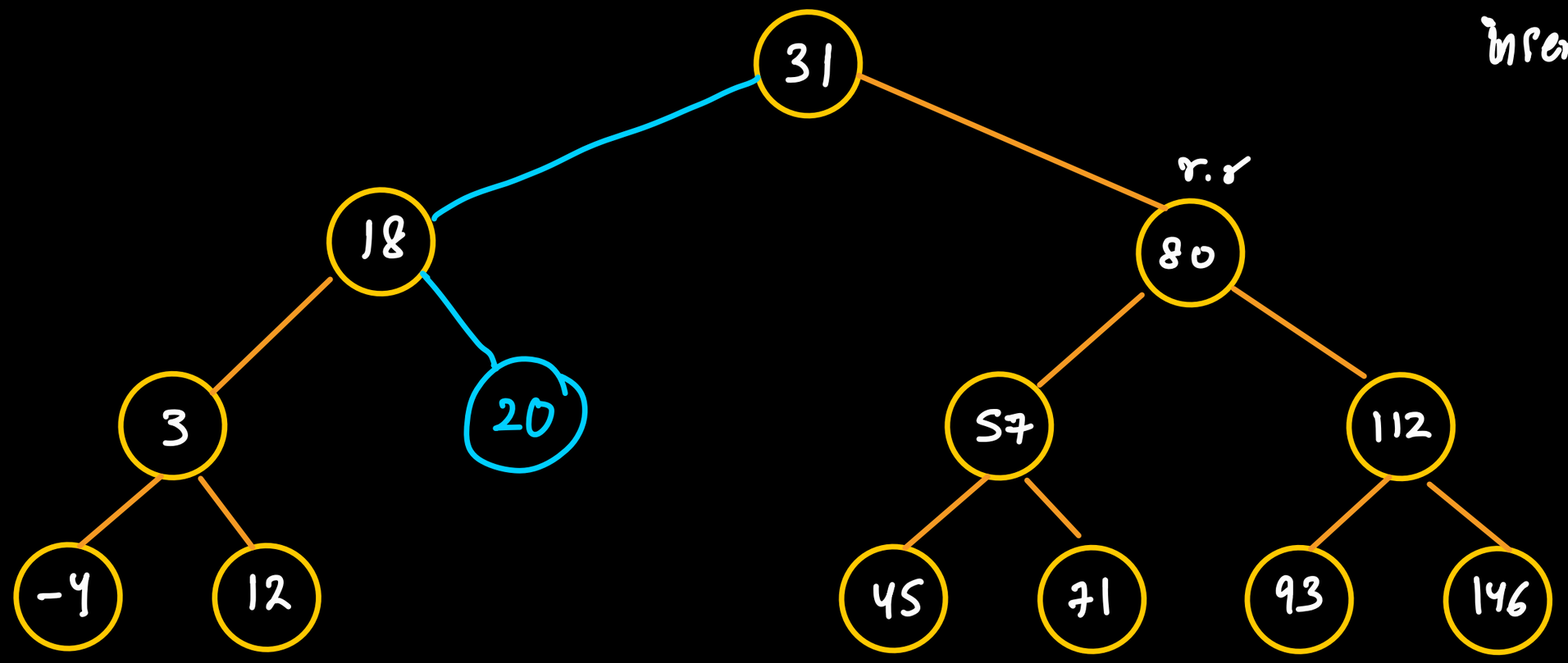
[LC 938]

Ques: Count BST nodes that lie in a given range

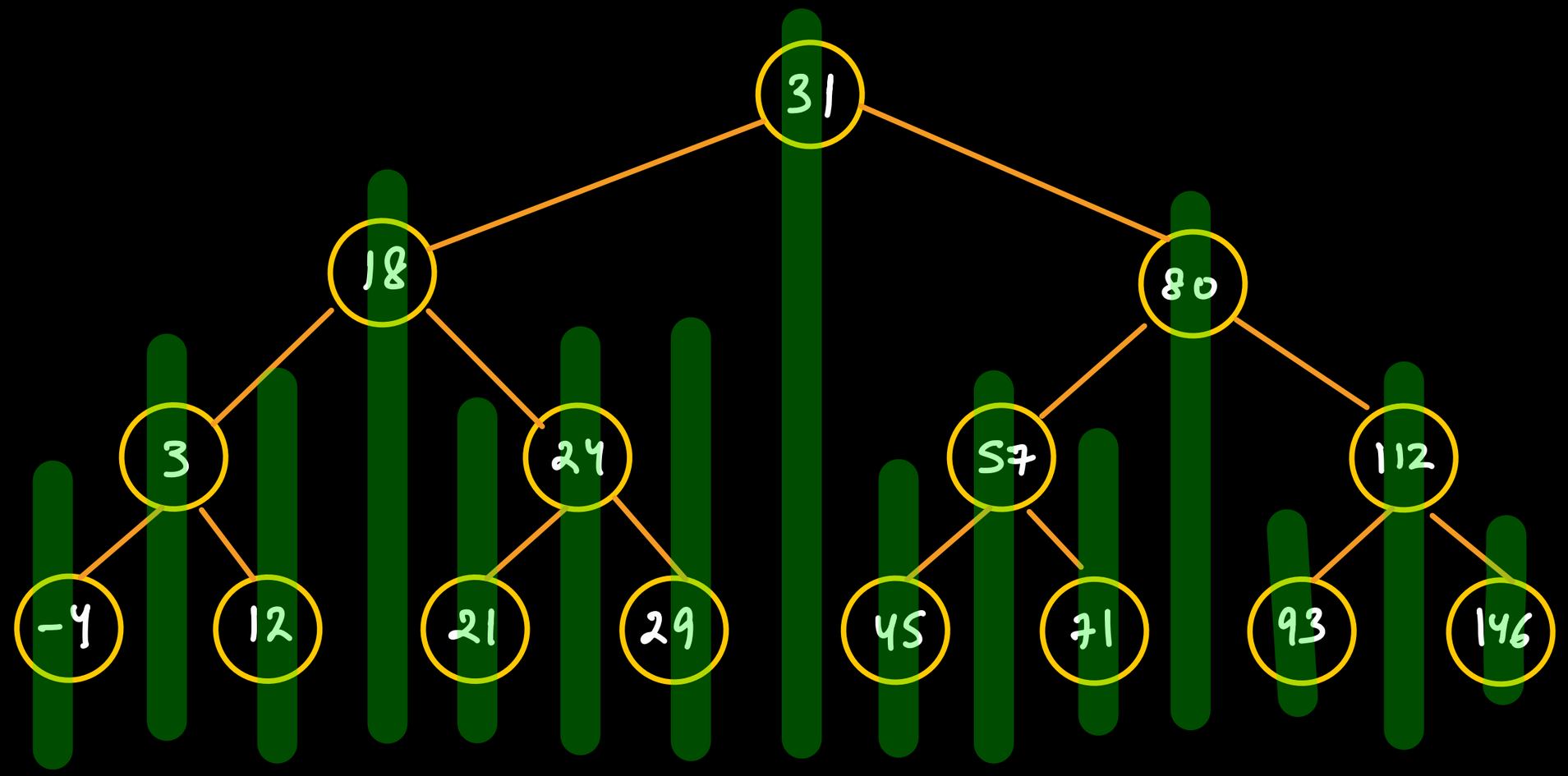
→ go to appropriate leaf node & then attach

Ques: Insert in a Binary Search Tree

insert (20)



Inorder Traversal of BST is Sorted



Ques: kth smallest element in a BST

M-1 find inorder of BST & store it in array, return arr[k-1]

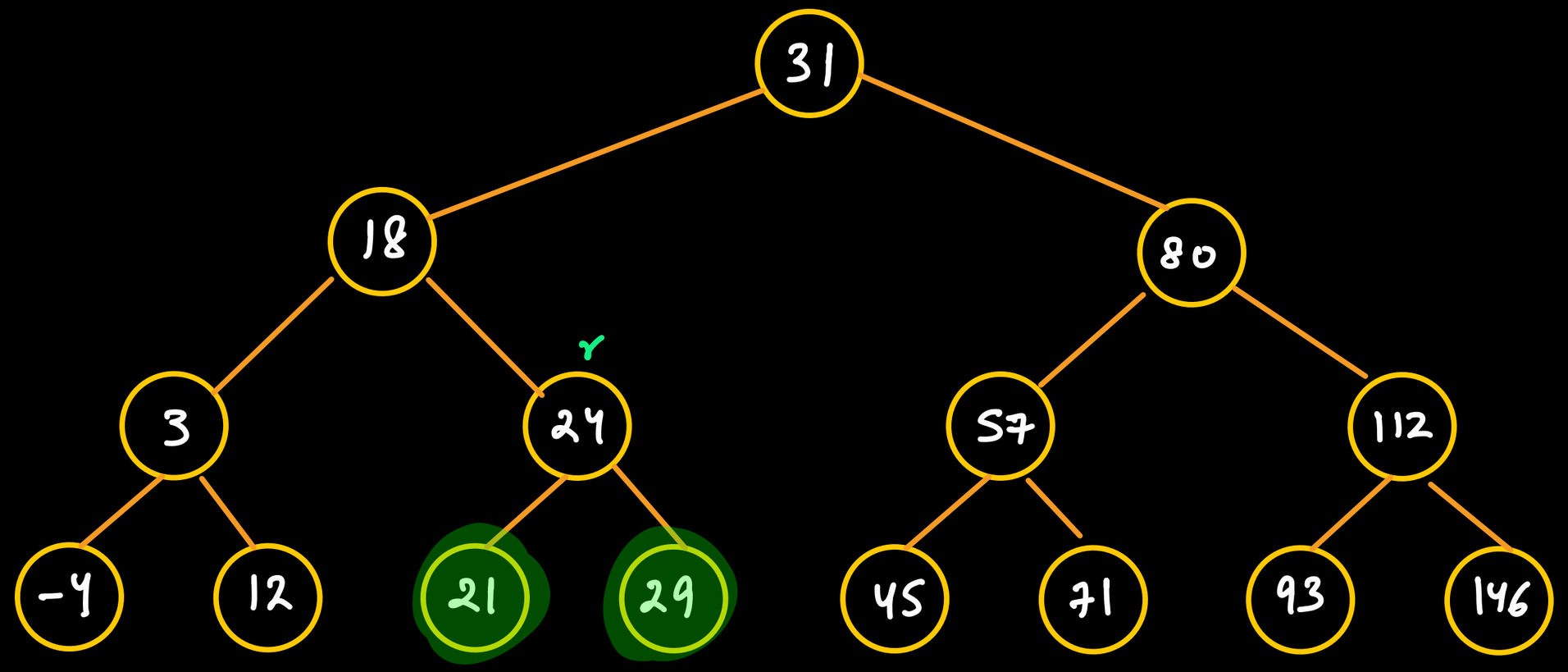
T.C. = O(n)

A.S. = O(n)

HW : Validate BST

Ques: LCA of 2 Nodes in a BST

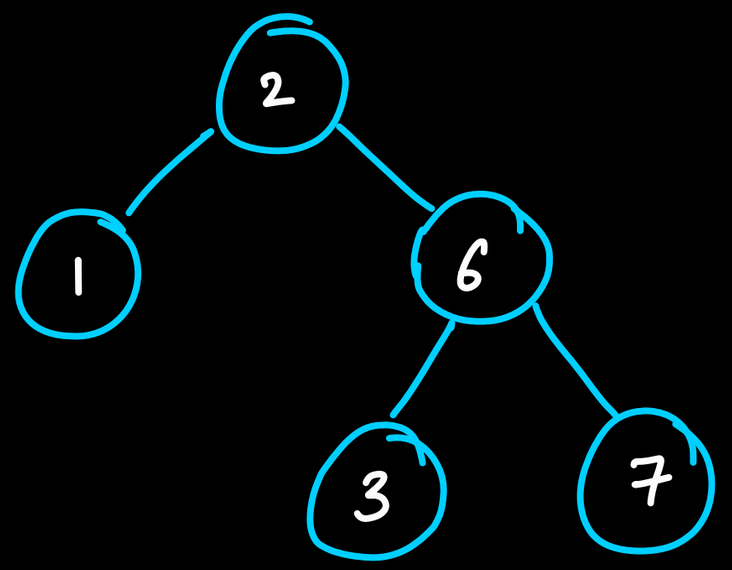
$p=21,\ q=29$



Ques: BST to GST

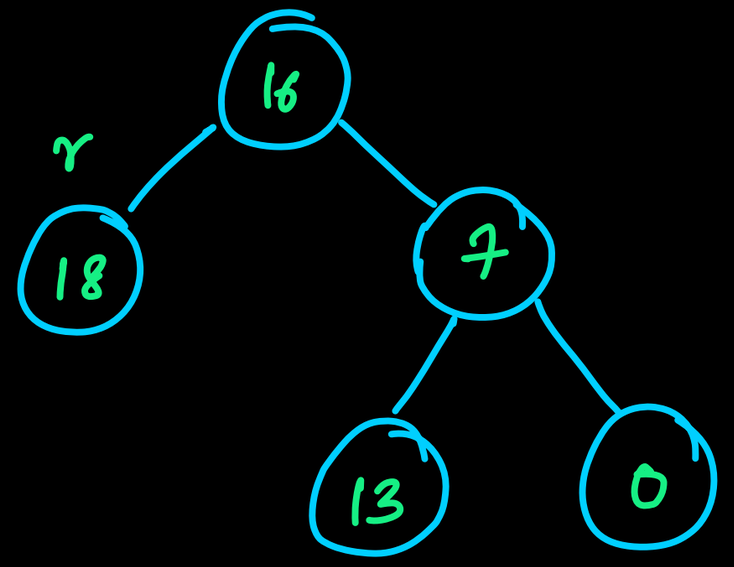
sum = ~~0~~ ~~7~~ ~~13~~ ~~16~~ ~~18~~ 19

val = ~~7~~
~~6~~
~~3~~
~~2~~
1



original

⇒



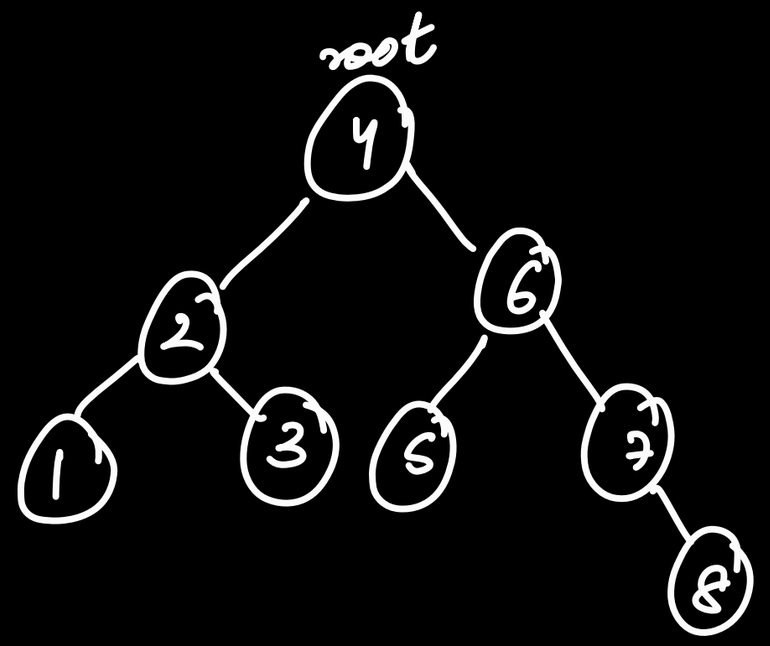

```
r in (root)
    base case
    rin(root.right)
    Kaam
    rin(root.left)
```

3

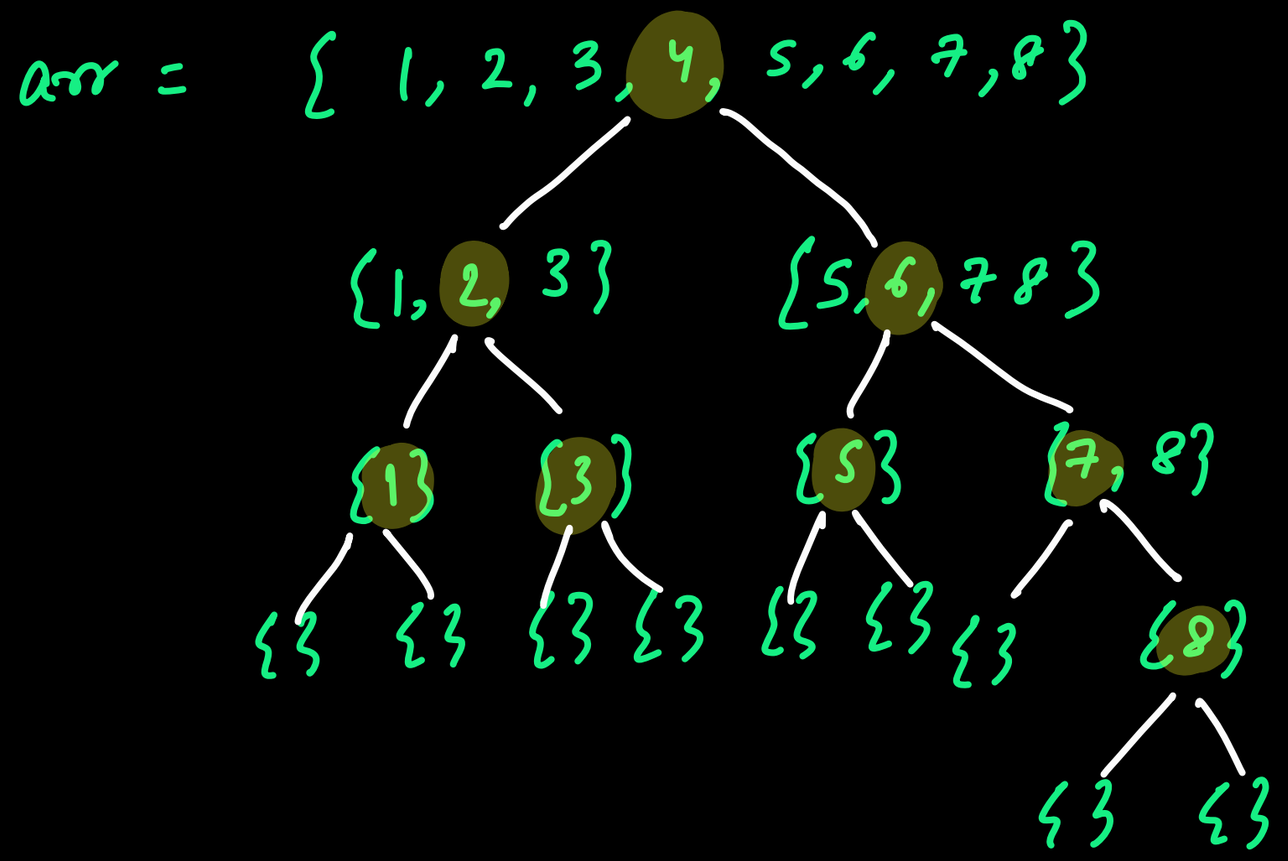
Ques: Convert sorted array to balanced BST

arr = { 1, 2, 3, 4, 5, 6, 7, 8 }

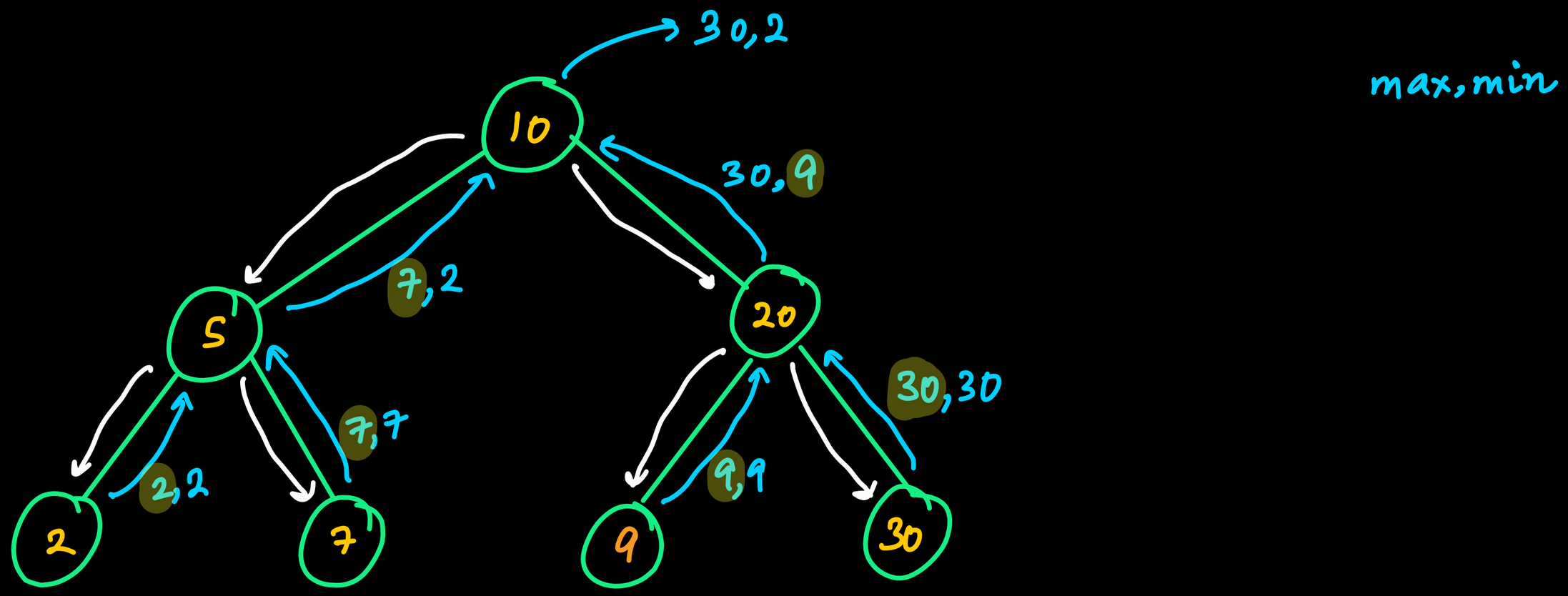
result



Ques: Convert sorted array to BST



Ques: Validate BST (max - min Pair return)



Ques: Largest BST

H.W. Max Sum BST in BT LC - 1373